



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO DD FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

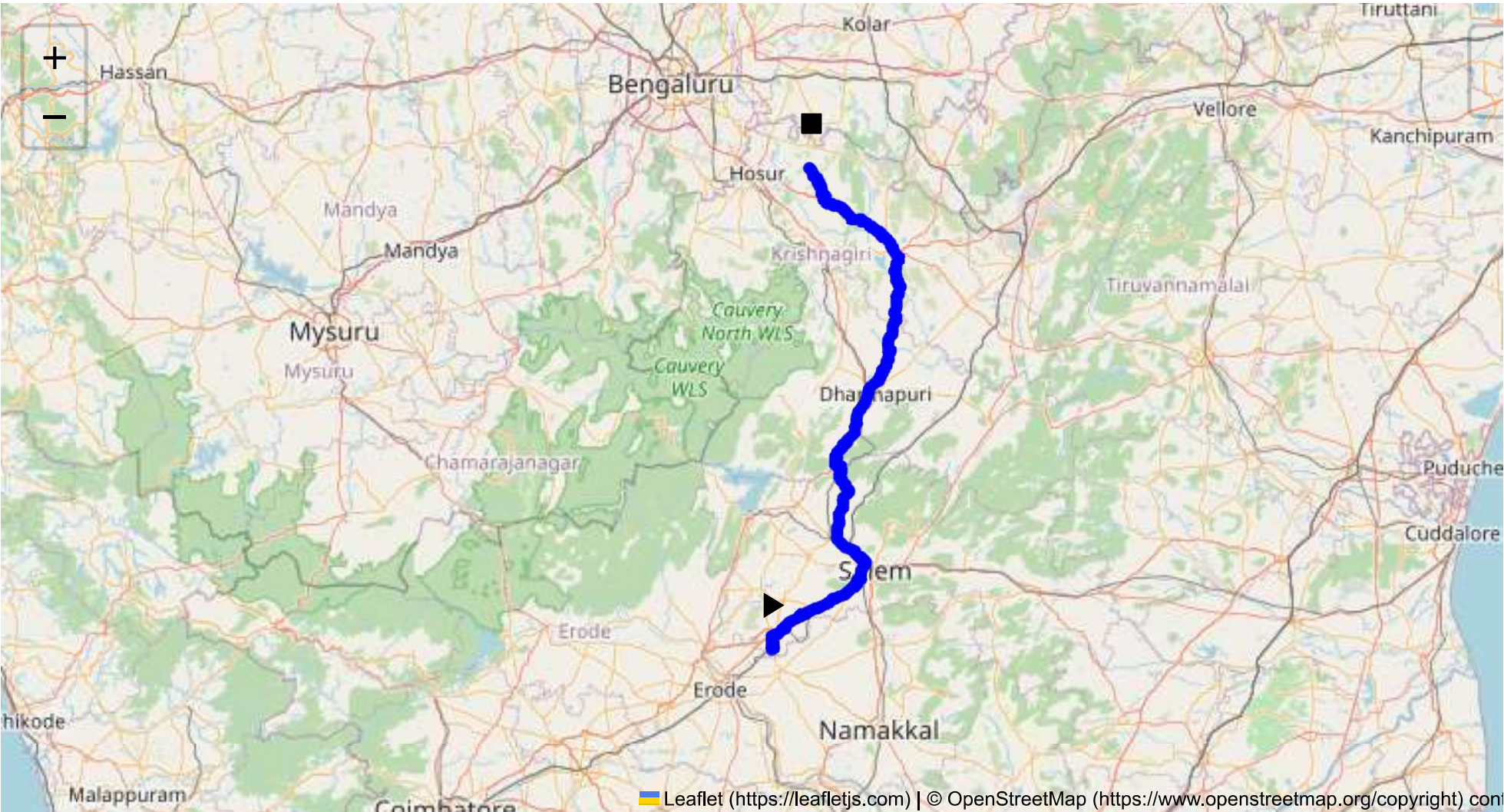
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 191.39 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours
Start: (11.4381, 77.8734)
End: (12.750149, 77.975897)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The journey from Sangagiri, Tamil Nadu to Athimugam, Tamil Nadu covers approximately 191.39 kilometers, primarily along National Highways and state roads. The route typically involves the NH44 and several smaller state roads that connect major truck routes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** Tamil Nadu experiences a tropical climate with high temperatures year-round. The monsoon season occurs from October to December, which can bring heavy rains.
- **Hazards:** During the monsoon, road conditions can deteriorate due to flooding, landslides in hilly areas, and reduced visibility, increasing the risk of accidents.

3. Analysis of Traffic Patterns

- **Peak Hours:** Peak traffic is usually observed in the early morning (7-9 AM) and late evening (5-8 PM) due to commuter and local business traffic.
- **Congestion-Prone Areas:** Hosur's entry points are known for congestion, particularly at industrial areas and markets along NH44.

4. Assessment of Road Quality and Infrastructure

- Major highways like NH44 are generally well-maintained, but smaller state roads may have potholes and uneven surfaces, especially after monsoon seasons.
- Frequent road maintenance and construction work could lead to temporary detours and disruptions.

5. Suggestions for Alternative Routes

- In case of emergencies or road blockages, using the NH79 to reach NH44 offers an alternative, albeit possibly longer, path as it connects various critical junctions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials requires special permits in Tamil Nadu.
- Regulations may restrict passage through ecologically sensitive zones and urban areas during peak hours.

7. Overview of Historical Incidents

- There have been reported incidents related to chemical spills and vehicular fires on NH44, especially when passing through densely populated regions near urban centers.

8. Environmental Considerations and Sensitive Areas

- Several zones along the route, such as river crossings and forest reserves, necessitate careful driving and adherence to environmental protection protocols.

9. Analysis of Communication Coverage

- Major highways like NH44 generally have good mobile network coverage.
- Potential communication dead zones are likely along some interior state roads and rural stretches.

10. Estimated Emergency Response Times

- Near urban centers like Hosur, emergency services can reach within 15-30 minutes.
- In rural or remote areas, response times could extend to 1 hour or more due to distance and road accessibility.

11. Overall Summary of Risk Assessment

- **Strengths:** Major highways facilitate quicker travel and better road conditions.
- **Weaknesses:** Smaller state roads can pose hazards due to poor maintenance and increased traffic during monsoon or festivals.
- **Risks:** Weather-related events and historical incidents necessitate heightened caution for hazardous material transport.
- **Recommendations:** Constant weather monitoring, adherence to transport regulations, and alternative route planning are essential for safe travel. Additionally, maintaining communication equipment to navigate potential dead zones is critical.

Overall, while the route provides efficient access between key locations, drivers must remain vigilant regarding weather impacts, road conditions, and traffic regulations to mitigate risks effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	Medium	11.43995, 77.87431	30 KM/Hr
2	Turn	High	11.44107, 77.87528	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.46400, 77.87323	30 KM/Hr
6	Turn	Medium	11.46765, 77.87037	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47986, 77.88397	30 KM/Hr
9	Turn	High	11.53606, 77.95869	15 KM/Hr
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	Medium	11.62932, 78.11659	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.64183, 78.11810	15 KM/Hr
13	Turn	Medium	11.64338, 78.11998	30 KM/Hr
14	Turn	Medium	11.67903, 78.12796	30 KM/Hr
15	Turn	Medium	11.69532, 78.10764	30 KM/Hr
16	Turn	Medium	11.73968, 78.05098	30 KM/Hr
17	Turn	Medium	11.86790, 78.08347	30 KM/Hr
18	Turn	Medium	11.91357, 78.05868	30 KM/Hr
19	Turn	Medium	11.92556, 78.06499	30 KM/Hr
20	Turn	Medium	11.93296, 78.06432	30 KM/Hr
21	Turn	Medium	11.93535, 78.06150	30 KM/Hr
22	Turn	Medium	11.93694, 78.05425	30 KM/Hr
23	Turn	Medium	11.95531, 78.04698	30 KM/Hr
24	Turn	High	11.95803, 78.04964	15 KM/Hr
25	Turn	Medium	11.96068, 78.04960	30 KM/Hr
26	Turn	High	11.96370, 78.05142	15 KM/Hr
27	Turn	High	11.96650, 78.04927	15 KM/Hr
28	Turn	Medium	12.22422, 78.19127	30 KM/Hr
29	Turn	Medium	12.24685, 78.19561	30 KM/Hr
30	Turn	Medium	12.25012, 78.19946	30 KM/Hr
31	Turn	Medium	12.25280, 78.20041	30 KM/Hr
32	Turn	Medium	12.29065, 78.19892	30 KM/Hr
33	Turn	Medium	12.32012, 78.20734	30 KM/Hr
34	Turn	Medium	12.40099, 78.21805	30 KM/Hr
35	Turn	Medium	12.45136, 78.21799	30 KM/Hr
36	Turn	Medium	12.50485, 78.22336	30 KM/Hr
37	Turn	Medium	12.59710, 78.13638	30 KM/Hr
38	Turn	Medium	12.61895, 78.08313	30 KM/Hr
39	Turn	Medium	12.63692, 78.07068	30 KM/Hr
40	Turn	Medium	12.66132, 78.01539	30 KM/Hr
41	Turn	High	12.67002, 78.01671	15 KM/Hr
42	Turn	Medium	12.67454, 78.01437	30 KM/Hr
43	Turn	High	12.67572, 78.01196	15 KM/Hr
44	Turn	Medium	12.70398, 78.00088	30 KM/Hr
45	Turn	High	12.71385, 77.99942	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	12.71574, 77.99671	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
17	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
18	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
19	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
21	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
22	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
24	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
25	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
26	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
35	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
36	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
38	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
43	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
44	hospital	St. Mary's Hospital	12.5128555, 78.2186887	30 km/h	Medium
45	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
46	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
47	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
48	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium
54	hospital	Shoolagiri Government Hospital	12.6669051, 78.015584	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
11	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
15	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
16	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
33	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
34	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
37	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
39	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
40	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
41	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
42	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
49	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium
50	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
51	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium